Rotary Positional Encoding: A Mathematical Summary

Relative Position, Matrix Exponentials, Generators, and Long-Context Behavior

Abstract

Rotary Position Embedding (RoPE) injects relative positional information into transformer attention by rotating query and key vectors in two-dimensional subspaces. The central mathematical structure is a one-parameter family of linear transformations $A(t)$ satisfying $A(s+t) = A(s)A(t)$, which under standard regularity assumptions can be written as a matrix exponential $A(t) = \mathrm{e}^{tX}$. For RoPE, the generator X is block diagonal with skew-symmetric 2×2 blocks, so the exponential becomes a collection of rotations at different frequencies. This note develops the connection from ordinary attention to relative rotations, explains why diagonalization is useful for classifying positional encodings, derives a natural positional similarity kernel, and discusses the implications of periodic phase structure for very long contexts.

1. Why Attention Needs Position

For a query vector $q_s \in \mathbb{R}^d$ at position s and a key vector $k_t \in \mathbb{R}^d$ at position t, the unnormalized attention logit is

$$\ell_{st} = \frac{q_s^\mathsf{T} k_t}{\sqrt{d}}. \tag{1}$$

The dot product measures compatibility between the learned query and key representations, but by itself it contains no information about where either token occurs in the sequence. If identical content vectors were placed at two different positions, the raw dot product would not distinguish those positions.

A positional encoding therefore modifies the query and key before their dot product is computed. In a general linear form,

$$Q_s = F(s)q_s, \qquad K_t = G(t)k_t. \tag{2}$$

The resulting score is

$$Q_s^\mathsf{T} K_t = q_s^\mathsf{T} F(s)^\mathsf{T} G(t) k_t. \tag{3}$$

Thus, the positional effect on attention is carried by the matrix

$$F(s)^\mathsf{T} G(t). \tag{4}$$

A central design goal is for this positional interaction to depend primarily on the relative displacement $\Delta = t - s$, rather than on the absolute positions s and t separately.

2. Relative Position and the Group Property

Suppose $A(t)$ denotes the linear transformation associated with moving by positional amount t. A natural translation-consistency requirement is

$$A(s+t) = A(s)A(t). \tag{5}$$

This equation states that moving by s and then by t should be equivalent to moving by $s+t$. It also implies

$$A(0) = \mathbf{I}, \qquad A(-t) = A(t)^{-1}. \tag{6}$$

Such a family is called a one-parameter group of linear transformations.

Under continuity or differentiability assumptions, every such family can be written as

$$\boxed{A(t) = \mathrm{e}^{tX}}, \tag{7}$$

where X is a fixed matrix called the *generator*. The generator can be recovered from the infinitesimal change at the origin:

$$X = A'(0). \tag{8}$$

The matrix exponential is defined by the power series

$$\mathrm{e}^{tX} = \mathbf{I} + tX + \frac{t^2X^2}{2!} + \frac{t^3X^3}{3!} + \cdots. \tag{9}$$

This is exactly the matrix analogue of the scalar exponential solution to $x'(t) = ax(t)$.

3. The Differential-Equation View

Consider the two-dimensional system

$$\frac{d}{dt}\begin{bmatrix} x(t) \\ y(t) \end{bmatrix} = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}\begin{bmatrix} x(t) \\ y(t) \end{bmatrix}. \tag{10}$$

The coefficient matrix

$$J = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix} \tag{11}$$

satisfies $J^2 = -\mathbf{I}$. Therefore,

$$\mathrm{e}^{tJ} = \mathbf{I}\cos t + J\sin t \tag{12}$$

$$= \begin{bmatrix} \cos t & -\sin t \\ \sin t & \cos t \end{bmatrix}. \tag{13}$$

Exponentiating this generator produces a rotation. This is the core algebraic mechanism used by RoPE.

More generally, introducing an angular frequency θ gives

$$X_\theta = \begin{bmatrix} 0 & -\theta \\ \theta & 0 \end{bmatrix}, \tag{14}$$

so that

$$\mathrm{e}^{tX_\theta} = R(t\theta) = \begin{bmatrix} \cos(t\theta) & -\sin(t\theta) \\ \sin(t\theta) & \cos(t\theta) \end{bmatrix}. \tag{15}$$

The generator answers the infinitesimal question: *what transformation should occur when position advances by a very small amount?*

4. RoPE as Relative Rotation

RoPE divides each attention-head representation into two-dimensional coordinate pairs. For the ith pair, let the assigned angular frequency be θ_i. At token position t, that pair is transformed by

$$R(t\theta_i). \tag{16}$$

For a full head with $m = d/2$ pairs, the positional transform is block diagonal:

$$R_t = \mathrm{diag}(R(t\theta_1), R(t\theta_2), \ldots, R(t\theta_m)). \tag{17}$$

Queries and keys become

$$q'_s = R_s q_s, \qquad k'_t = R_t k_t. \tag{18}$$

Their dot product is

$$(q'_s)^\mathsf{T} k'_t = q_s^\mathsf{T} R_s^\mathsf{T} R_t k_t \tag{19}$$

$$= q_s^\mathsf{T} R_{t-s} k_t. \tag{20}$$

The second equality follows because each rotation satisfies

$$R(a)^\mathsf{T} = R(-a), \qquad R(-a)R(b) = R(b-a). \tag{21}$$

Hence the positional interaction depends on the relative displacement

$$\boxed{\Delta = t - s} \tag{22}$$

rather than on s and t independently.

5. Geometric Interpretation of Rotation in Attention

For ordinary vectors q and k,

$$q^\mathsf{T} k = \|q\|\|k\| \cos\alpha, \tag{23}$$

where α is the angle between them. A rotation therefore changes the dot product by changing angular alignment while preserving norm.

For a single two-dimensional RoPE pair, write

$$q_i = r_q \begin{bmatrix} \cos\alpha \\ \sin\alpha \end{bmatrix}, \qquad k_i = r_k \begin{bmatrix} \cos\beta \\ \sin\beta \end{bmatrix}. \tag{24}$$

After introducing relative positional rotation, the contribution of this pair becomes

$$\boxed{q_i^\mathsf{T} R(\Delta\theta_i) k_i = r_q r_k \cos(\beta - \alpha + \Delta\theta_i)}. \tag{25}$$

This equation separates two geometric effects:

$$\underbrace{\beta - \alpha}_{\text{content geometry}} \qquad + \qquad \underbrace{\Delta\theta_i}_{\text{position geometry}} \quad . \tag{26}$$

The rotation does not itself assign a probability to a position. Instead, it changes the query-key logit. The attention probability is produced only after all logits are normalized by softmax:

$$p(t \mid s) = \frac{\exp(\ell_{st})}{\sum_j \exp(\ell_{sj})}, \tag{27}$$

where

$$\ell_{st} = \frac{q_s^\top R_{t-s} k_t}{\sqrt{d}}. \tag{28}$$

Thus the computation proceeds as

$$\text{relative position} \longrightarrow \text{rotation} \longrightarrow \text{dot product} \longrightarrow \text{logit} \longrightarrow \text{softmax probability}. \tag{29}$$

6. Why RoPE Uses Multiple Frequencies

A single rotation frequency is periodic. For one block,

$$R(\Delta\theta) = R(\Delta\theta + 2\pi n), \qquad n \in \mathbb{Z}. \tag{30}$$

Its positional period is therefore

$$\boxed{T = \frac{2\pi}{\theta}}. \tag{31}$$

If only one frequency were used, sufficiently separated relative positions could produce identical rotations.

RoPE instead uses many frequencies $\theta_1, \ldots, \theta_m$. The relative position Δ is therefore represented by a phase fingerprint

$$\boxed{\Phi(\Delta) = \left(\mathrm{e}^{i\Delta\theta_1}, \mathrm{e}^{i\Delta\theta_2}, \ldots, \mathrm{e}^{i\Delta\theta_m}\right).} \tag{32}$$

A high frequency changes phase rapidly and is sensitive to small positional changes. A low frequency changes slowly and carries coarse-scale positional information. The use of many scales makes it much harder for two different displacements to produce the same joint phase configuration.

In standard RoPE, frequencies are commonly spread geometrically. A schematic form is

$$\theta_i = B^{-2i/d}, \tag{33}$$

for a base B and feature-pair index i. The important qualitative fact is the logarithmic spread of timescales: some dimensions rotate rapidly, while others rotate slowly.

7. A Positional Similarity Kernel

A useful way to study positional distinguishability independently of token content is to average the phase alignment across RoPE dimensions. Define

$$\boxed{C(\Delta) = \frac{1}{m} \sum_{i=1}^{m} \cos(\Delta\theta_i)}. \tag{34}$$

This quantity is not a Pearson correlation coefficient; it is better interpreted as a normalized positional similarity kernel. If

$$C(\Delta) \approx 1, \tag{35}$$

then many RoPE blocks are close to their original orientation for that displacement. If $C(\Delta)$ is near zero, the phases are broadly decorrelated.

For a context window of length L, the maximum relative displacement satisfies

$$|\Delta| \leq L - 1. \tag{36}$$

The approximate number of complete rotations made by frequency θ_i across the context is

$$\boxed{N_i(L) \approx \frac{L\theta_i}{2\pi}.} \tag{37}$$

This provides a simple way to relate a frequency spectrum to a target context length.

8. Long-Context Behavior and Phase Aliasing

Because every individual RoPE component is periodic, each phase eventually wraps around. For a single frequency, two displacements differ by an exact positional period whenever

$$(\Delta_1 - \Delta_2)\theta = 2\pi n. \tag{38}$$

With many frequencies, a full collision would require

$$(\Delta_1 - \Delta_2)\theta_i \approx 2\pi n_i \qquad \text{for all } i, \tag{39}$$

which is much more restrictive.

Nevertheless, pure RoPE does not create an infinitely unique coordinate system. It creates a high-dimensional quasi-periodic phase code. At sufficiently extreme displacements, near-recurrences can occur, and all individual phase components may have wrapped many times. This is one reason very long-context extrapolation is nontrivial.

A second and often more practical issue is distribution shift. A RoPE formula may be mathematically defined for any integer position, but a model trained only on

$$0 \leq \Delta \leq L_{\text{train}} \tag{40}$$

may not have learned to interpret phase combinations encountered at

$$\Delta \gg L_{\text{train}}. \tag{41}$$

Thus one should distinguish

$$\text{mathematical definability} \qquad \text{from} \qquad \text{learned long-context generalization.} \tag{42}$$

Unlike a probability density, token position is not random in the RoPE construction. It is a deterministic index

$$t \in \{0, 1, 2, \ldots, L - 1\} \tag{43}$$

mapped to a collection of phases. Probability enters only later through the attention softmax.

9. Why Diagonalize the Generator?

Diagonalization is not required to execute RoPE. Its role is primarily analytical: it reveals the independent modes hidden inside a general generator.

Suppose

$$X = PDP^{-1}, \tag{44}$$

where

$$D = \mathrm{diag}(\lambda_1, \ldots, \lambda_d). \tag{45}$$

Then

$$\mathrm{e}^{tX} = P\mathrm{e}^{tD}P^{-1}, \tag{46}$$

with

$$\mathrm{e}^{tD} = \mathrm{diag}(\mathrm{e}^{t\lambda_1}, \ldots, \mathrm{e}^{t\lambda_d}). \tag{47}$$

This converts a coupled dynamical system into independent modes.

If an eigenvalue is

$$\lambda = u + i\theta, \tag{48}$$

then

$$\mathrm{e}^{t\lambda} = \mathrm{e}^{ut}\mathrm{e}^{i\theta t}. \tag{49}$$

The real part u controls exponential growth or decay, while the imaginary part θ controls oscillation or rotation. In real two-dimensional coordinates, this mode has the form

$$\mathrm{e}^{ut}\begin{bmatrix}\cos(\theta t) & -\sin(\theta t)\\ \sin(\theta t) & \cos(\theta t)\end{bmatrix}. \tag{50}$$

Thus, for diagonalizable generators, positional dynamics can be understood as combinations of scaling/decay and rotation.

RoPE chooses the special case

$$u = 0, \tag{51}$$

so that the positional transformation preserves norm and consists only of rotations.

10. Choosing a Generator

There is no single mathematically optimal generator independent of the task and training distribution. The theory narrows the design space by identifying interpretable families of generators. For RoPE, the generator is chosen to be block diagonal:

$$X = \mathrm{diag}\left(\begin{bmatrix}0 & -\theta_1\\ \theta_1 & 0\end{bmatrix}, \ldots, \begin{bmatrix}0 & -\theta_m\\ \theta_m & 0\end{bmatrix}\right). \tag{52}$$

This structure has several useful properties:

- It yields an exact relative-position identity in the query-key dot product.
- It preserves vector norms because each block is orthogonal after exponentiation.
- It avoids exponential growth or shrinkage because the eigenvalues are purely imaginary.
- It reduces a large matrix-design problem to choosing a spectrum of frequencies $\{\theta_i\}$.

From an optimization viewpoint, an "optimal" frequency spectrum would depend on quantities such as training loss, expected dependency lengths, context-window size, retrieval quality, numerical stability, and extrapolation behavior. Long-context RoPE variants can therefore be viewed as modifications to the effective frequency spectrum so that the phase geometry remains useful over a larger interval of Δ.

11. Decay, Rotation, and Other Generator Classes

The eigenvalue view provides a compact taxonomy. If

$$\lambda = u + i\theta, \tag{53}$$

then:

Eigenvalue structure	Behavior	Interpretation
$\lambda = 0$	constant	no positional evolution
$u < 0,\ \theta = 0$	exponential decay	recency-like attenuation
$u = 0,\ \theta \neq 0$	pure rotation	RoPE-like phase encoding
$u < 0,\ \theta \neq 0$	decaying rotation	phase plus attenuation

Non-diagonalizable generators introduce polynomial factors through Jordan blocks. For example,

$$X = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix} \qquad \Longrightarrow \qquad X^2 = 0, \tag{54}$$

so

$$\mathrm{e}^{tX} = \mathbf{I} + tX = \begin{bmatrix} 1 & t \\ 0 & 1 \end{bmatrix}. \tag{55}$$

This creates linear positional dependence rather than periodic rotation. Such polynomial terms help connect the generator framework to linear positional-bias schemes such as ALiBi.

12. Core Mental Model

The most compact way to think about RoPE is:

1. Content determines the learned query and key vectors.
2. Position rotates each two-dimensional coordinate pair at a prescribed frequency.
3. The relative rotation between positions s and t depends only on $\Delta = t - s$.
4. The dot product combines content geometry with positional phase.
5. Softmax converts the resulting logits into attention probabilities.

Equivalently, RoPE represents displacement through a phase fingerprint

$$\Phi(\Delta) = \left(\mathrm{e}^{i\Delta\theta_1}, \ldots, \mathrm{e}^{i\Delta\theta_m}\right), \tag{56}$$

and the transformer learns how those relative phases should modulate content-based attention.

13. Key Equations

$$A(t) = \mathrm{e}^{tX}, \tag{57}$$

$$R(t\theta) = \begin{bmatrix} \cos(t\theta) & -\sin(t\theta) \\ \sin(t\theta) & \cos(t\theta) \end{bmatrix}, \tag{58}$$

$$(R_s q_s)^{\mathsf{T}} (R_t k_t) = q_s^{\mathsf{T}} R_{t-s} k_t, \tag{59}$$

$$\ell_{st} = \frac{q_s^{\mathsf{T}} R_{t-s} k_t}{\sqrt{d}}, \tag{60}$$

$$p(t \mid s) = \frac{\exp(\ell_{st})}{\sum_j \exp(\ell_{sj})}, \tag{61}$$

$$T_i = \frac{2\pi}{\theta_i}, \tag{62}$$

$$C(\Delta) = \frac{1}{m} \sum_{i=1}^{m} \cos(\Delta\theta_i), \tag{63}$$

$$N_i(L) \approx \frac{L\theta_i}{2\pi}. \tag{64}$$

References

[1] A. Vaswani et al., "Attention Is All You Need," *Advances in Neural Information Processing Systems*, 2017.

[2] J. Su et al., "RoFormer: Enhanced Transformer with Rotary Position Embedding," arXiv:2104.09864, 2021; later published in *Neurocomputing*.

[3] O. Press, N. A. Smith, and M. Lewis, "Train Short, Test Long: Attention with Linear Biases Enables Input Length Extrapolation," arXiv:2108.12409, 2021.

[4] Jane Street Tech Blog, "Using Group Theory to Explore the Space of Positional Encodings for Attention."